

LAC BENOIT, QC, Canada

WMO: 715200

Lat: 51.532N

Lon: 71.111W

Elev: 1801

StdP: 13.76

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.6	-36.6	-49.1	0.3	-41.8	-42.9	0.5	-36.3	27.8	9.9	24.2	9.2	1.4	150	0.788

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	78.5	62.3	75.0	60.6	71.7	59.7	66.1	72.4	64.1	70.1	62.1	67.9	9.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	96.2	68.0	62.0	88.9	65.9	60.0	82.5	64.3	31.9	72.1	30.3	70.3	28.8	67.7	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.6	20.9	18.8	-51.7	85.5	4.1	3.1	-54.7	87.7	-57.1	89.5	-59.4	91.2	-62.4	93.5		
			-50.2	69.9	3.1	1.8	-52.4	71.2	-54.3	72.3	-56.0	73.3	-58.3	74.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	-6.2	-4.5	9.3	26.3	41.1	53.2	58.3	56.2	47.2	34.8	20.0	2.6	(6)
	DBStd	25.21	15.18	14.97	15.33	10.73	8.04	7.80	5.64	5.92	7.33	7.65	10.84	14.38	(7)
	HDD50	8581	1741	1526	1261	710	294	52	4	12	139	472	900	1470	(8)
	HDD65	13392	2206	1946	1726	1160	741	361	220	278	534	935	1350	1935	(9)
	CDD50	691	0	0	0	0	19	149	260	205	56	2	0	0	(10)
	CDD65	25	0	0	0	0	0	7	11	6	1	0	0	0	(11)
	CDH74	405	0	0	0	0	18	128	169	79	11	0	0	0	(12)
	CDH80	56	0	0	0	0	1	23	23	8	1	0	0	0	(13)
(14) Wind	WSAvg	8.3	8.1	7.4	8.1	8.4	7.9	8.1	8.4	8.3	9.1	9.5	9.2	7.6	(14)
(15) Precipitation	PrecAvg	36.3	1.9	1.6	1.9	2.4	2.8	3.6	4.9	4.6	4.3	3.2	3.0	2.2	(15)
	PrecMax	42.6	3.6	2.8	3.3	4.0	5.7	6.0	7.1	6.9	9.1	4.7	4.7	4.2	(16)
	PrecMin	29.9	0.9	0.5	0.6	0.7	1.4	1.5	2.6	2.3	1.9	1.9	1.8	1.2	(17)
	PrecStd	3.2	0.7	0.5	0.7	0.8	0.9	1.1	1.1	1.2	1.4	0.8	0.8	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	35.9	45.4	57.2	76.4	83.2	83.1	80.8	74.6	60.1	44.8	35.3	(19)
		MCWB	33.3	33.7	37.6	42.7	55.9	64.7	64.7	65.0	62.9	52.6	43.4	34.4	(20)
	2%	DB	29.4	31.3	39.6	50.6	68.5	77.5	78.6	75.9	68.8	54.1	39.6	32.3	(21)
		MCWB	29.0	29.1	34.0	39.9	53.2	60.6	62.7	62.7	60.0	49.9	38.8	31.9	(22)
	5%	DB	22.5	25.5	35.6	45.9	62.9	73.4	75.0	71.8	64.2	50.4	35.8	28.0	(23)
		MCWB	21.9	24.0	31.6	37.6	50.4	58.9	61.2	61.4	58.2	47.7	35.1	27.5	(24)
	10%	DB	17.3	20.1	32.3	41.9	57.6	69.4	71.4	68.2	60.1	46.7	33.5	24.0	(25)
		MCWB	16.6	18.6	29.6	35.2	47.7	56.6	60.1	59.8	55.4	44.4	32.9	23.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.9	33.9	39.6	44.0	59.3	68.0	68.9	68.3	65.5	55.3	43.9	34.8	(27)
		MCDB	33.1	35.5	42.8	53.9	68.6	76.0	76.1	74.2	69.3	57.8	44.1	35.1	(28)
	2%	WB	28.8	29.2	35.4	40.7	55.7	64.6	66.1	65.6	62.1	51.3	38.5	32.2	(29)
		MCDB	29.1	31.3	38.4	48.5	64.6	72.1	72.3	71.4	66.7	53.2	38.8	32.5	(30)
	5%	WB	22.0	24.0	32.4	38.6	52.5	61.4	63.9	63.6	59.2	48.5	34.9	27.7	(31)
		MCDB	22.5	25.7	35.0	44.3	60.9	68.7	70.8	68.4	62.7	50.5	35.3	28.0	(32)
	10%	WB	16.8	18.3	29.4	36.5	49.1	58.8	62.0	61.4	56.1	44.8	32.7	23.8	(33)
		MCDB	17.5	19.8	32.6	41.2	57.4	66.1	68.4	66.1	59.5	46.6	33.1	24.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	24.0	28.8	29.6	24.3	23.0	24.7	22.5	20.9	18.7	13.3	13.7	20.1	(35)
		MCDBR	20.5	22.3	27.1	28.4	34.5	32.3	30.0	27.2	25.6	19.7	12.1	15.2	(36)
	5% WB	MCWBR	20.0	20.5	21.9	19.0	20.4	18.0	16.3	15.9	16.8	15.4	11.3	14.3	(37)
		MCWBR	20.3	21.5	23.8	23.3	29.5	24.9	22.7	21.5	20.4	17.4	11.7	14.8	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.245	0.259	0.272	0.317	0.346	0.364	0.377	0.353	0.335	0.302	0.270	0.243	(40)
		taud	2.222	2.220	2.224	2.186	2.375	2.395	2.383	2.469	2.491	2.532	2.369	2.211	(41)
	Ebn at Noon	Ebn at Noon	247	277	295	292	288	282	276	278	271	262	238	226	(42)
		Edh at Noon	21	29	35	41	36	36	36	31	27	21	19	18	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	374	738	1194	1561	1726	1705	1525	1222	927	515	252	226	(44)
	RadStd	RadStd	31	55	92	114	112	140	131	83	69	52	37	36	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page